ANIS SUHANA BINTI MOHD AZMI

Penang, Malaysia | +6010 387 0486 | anisya020308@gmail.com | linkedin.com/in/anissuhana

PROFESSIONAL SUMMARY

Electronic Engineering graduate from Universiti Teknologi MARA (UiTM) with First Class Honours and hands-on ASIC Physical Design training at Penang Skills Development Centre (PSDC). Completed trainer-guided lab exercises covering ASIC design flow, logic synthesis, floorplanning, placement, clock tree synthesis (CTS), routing, timing analysis, and physical verification concepts using Synopsys EDA tools. Experienced in writing TCL scripts for training tasks and familiar with SDC constraints, power-grid planning, and UNIX/Linux environments. Seeking an entry-level Physical Design Engineer opportunity to contribute to CPU implementation and further develop skills in timing closure, power optimization, and physical verification.

TECHNICAL SKILLS

ASIC Physical Design: Floorplanning, Macro Placement, Power Grid Planning, Placement, Clock Tree Synthesis (CTS), Routing, RDL Routing, I/O Pad and Bump Planning, Utilization and Congestion Analysis

ASIC Design Flow: ASIC Design Flow, RTL Design Fundamentals, Logic Synthesis, SDC Design Constraints, Netlist-to-GDSII Flow Concepts

EDA Tools: Synopsys Design Compiler, Synopsys Formality, Synopsys Fusion Compiler, Synopsys PrimeTime, IC Validator, Gowin EDA, Proteus, LTspice, AutoCAD

Timing and Physical Verification: Static Timing Analysis (STA) Fundamentals, SDC Constraints, Clock Reports, Timing Checks, Timing Sign-Off Concepts, DRC/LVS Fundamentals, Formal Verification, UPF Power Intent

Programming and Scripting: TCL Scripting, Python, C, C++, MATLAB, LabVIEW

Computing Environments: UNIX, Linux, Ubuntu

Hardware and Embedded: Circuit Design, PCB Design, STM32, Arduino, Sensor Integration, LiDAR, Robot Operating System (ROS)

Industrial Automation: Siemens TIA Portal, SIMATIC Manager, Twinsoft

PROFESSIONAL EXPERIENCE

Physical Design Trainee, Penang Skills Development Centre (PSDC) **Sep 2026**

- Completed trainer-guided laboratory exercises covering ASIC design flow, RTL design, logic synthesis, and SDC design constraints.
- Performed hands-on training exercises using Synopsys Design Compiler, Formality, and Fusion Compiler across synthesis and physical implementation topics.
- Practised physical design stages including floorplanning, power-grid planning, placement, clock tree synthesis (CTS), and routing.
- Completed training modules on STA, timing checks, timing sign-off, DRC/LVS, and UPF power intent.
- Wrote TCL scripts for training tasks and studied RM flow and design-flow automation.
- Currently undertaking the Bitcoin Project, the final team project of the PSDC programme: running the Fusion Compiler flow from init_design to chip_finish on a Bitcoin mining design, analysing timing QoR, pushing toward higher frequency, and applying ECO with Formality verification.

Industrial Trainee, Tenaga Nasional Berhad (TNB) **Oct 2025**

- Interpreted as-built drawings for utility installation and removal work related to the TNB SAVR project.
- Developed familiarity with power equipment, including transformers, vacuum circuit breakers, and ring main units.
- Supported technical documentation, daily work logs, and Contractor Safety Quality Assurance (CSQA) records.

Industrial Trainee, Telekom Malaysia **Apr 2022**

- Assisted with telecommunications equipment installation, troubleshooting, and network maintenance.
- Coordinated with customers and technical teams to help resolve network issues.

PROJECTS

Full-Chip Floorplanning and ORCA_TOP Block Implementation, Physical Design **Sep 2026**

- Developed a full-chip floorplan by defining die dimensions, aspect ratio, I/O pad placement, and bump configuration according to project constraints.
- Planned power and ground pad distribution and power-grid connections to support chip-level power distribution requirements.
- Performed RDL routing for bump connections and designated clock nets, including differential routing for specified clock signals.
- Implemented the ORCA_TOP block floorplan using Synopsys Fusion Compiler, incorporating fixed macros, bottom-side pin placement, and power-grid planning.
- Reviewed utilization and congestion against the project requirements.

ANN-Based Diagnostic Model for Yellow-Flesh Watermelon Ripeness, Final Year Project **Jul 2026**

- Developed a MATLAB Artificial Neural Network classifier to grade watermelon ripeness using Visible (VIS) and Near-Infrared (NIR) spectral data.
- Compared three training functions across 14 hidden-neuron configurations using accuracy, sensitivity, and specificity.
- Achieved 83.3% accuracy, 86.7% sensitivity, and 80.0% specificity using the traingda algorithm.

Autonomous Mobile Robot Navigation Using LiDAR and ROS **Jun 2026**

- Configured Ubuntu Linux and ROS for autonomous navigation, mapping, and localization using LiDAR, Hector SLAM, and RViz.
- Developed a Python program using the Dynamic Window Approach (DWA) local planner to navigate to three destinations while avoiding obstacles.

SmartOptix: Eye Protection System (Arduino Uno, Buzzer, Blynk) **Oct 2024**

- Developed an Arduino-based prototype that monitored user proximity to a screen and triggered buzzer alerts and Blynk notifications.

Children Falling Prevention Balcony (STM32, Servo Motor, Ultrasonic Sensor, LCD) **May 2024**

- Built a prototype that detects distance with an ultrasonic sensor, closes access with a servo motor, and shows early warnings on an LCD.

Prevention House Box System (Arduino Uno, GSM Module, ESP Camera) **Oct 2022**

- Developed a home security system that detects intrusions and sends SMS and image notifications to users.

EDUCATION

Physical Design Program **Penang Skills Development Centre, Penang**

Bachelor of Electronic Engineering with Honours (First Class), CGPA 3.53 **UiTM Shah Alam, Selangor**

Dean's List for semesters 5, 6, 7 and 8

Diploma in Electrical Engineering (Electronics) **UiTM Pasir Gudang, Johor**

Dean's List for semesters 1, 3, 4 and 6

CERTIFICATIONS AND AWARDS

- Expert Craftsman in Mechatronics
- Siemens TIA Portal 1 and 2
- Omada Certified Network Administrator (OCNA) Wireless
- CA1C Customer Premises
- CA2C Network
- Construction Industry Development Board (CIDB Green Card)
- NIOSH TM Safety Passport (NTMSP)
- Gold Award, i-RiSE

ACTIVITIES AND LANGUAGES

Volunteer Student, IEEE Symposium on Industrial Electronics and Applications (ISIEA) **Jul 2026**

- Coordinated presenters and supported attendee registration and check-in.

Volunteer Student, IEEE Wireless Communications and Networking Conference (WCNC 2026) **Apr 2026**

- Assisted registration, guided presenters to session rooms, and gave on-site technical support in Kuala Lumpur.

Team Member, Urock Robotic 2024, PICC **Oct 2024**

- Worked with electronics, IT, and mechanical peers to meet project milestones.

Judge, Robothon Challenge 2024 **Jul 2024**

- Served on the judging panel, evaluating participants' robotic solutions and automation systems.
- Gave detailed evaluations and constructive feedback on the design, functionality, and usability of the robots.
- Collaborated with the organizing team to ensure competition standards were met and judging was fair and transparent.

Languages: Malay, English, Italian

REFERENCES

Dr. Noor Ezan Abdullah, Supervisor FYP, UiTM Shah Alam
Tel: 012 242 3192 | Email: nezan007@gmail.com